任意滤过复形的 K-INJECTIVE 兼容解消

Editorial reading note. Author text is retained verbatim. In the product over p-prime greater than p, the factor printed J^{p,bullet} is indexed by p-prime as in the surrounding displayed definitions. The composition on source line 2214 has the order of the two alpha factors reversed by type. Quotients F^p/F^{p-prime} use p-prime at least p. These are editorial reading notes, not changes to the author body.

Counting scope. The primary problem is Tag 0BKI, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Definition 0.1. Let $\mathcal{A}$ be an abelian category. A *filtered complex* $K^\bullet$ *of* $\mathcal{A}$ is a complex of $\mathrm{Fil}(\mathcal{A})$ (see Definition 0.2).

We will denote the filtration on the objects by F. Thus F^pK^n denotes the pth step in the filtration of the nth term of the complex. Note that each $F^pK^\bullet$ is a complex of $\mathcal{A}$. Hence we could also have defined a filtered complex as a filtered object in the (abelian) category of complexes of $\mathcal{A}$. In particular $\mathrm{gr}K^\bullet$ is a graded object of the category of complexes of $\mathcal{A}$.

Definition 0.2. Let $\mathcal{A}$ be an abelian category.

(1) A *decreasing filtration* F on an object A is a family $(F^nA)_{n\in\mathbf{Z}}$ of subobjects of A such that

$$A \supset \ldots \supset F^nA \supset F^{n+1}A \supset \ldots \supset 0$$

(2) A *filtered object of* $\mathcal{A}$ is pair (A, F) consisting of an object A of $\mathcal{A}$ and a decreasing filtration F on A.
(3) A *morphism* $(A, F) \to (B, F)$ *of filtered objects* is given by a morphism $\varphi : A \to B$ of $\mathcal{A}$ such that $\varphi(F^iA) \subset F^iB$ for all $i \in \mathbf{Z}$.
(4) The category of filtered objects is denoted $\mathrm{Fil}(\mathcal{A})$.
(5) Given a filtered object (A, F) and a subobject $X \subset A$ the *induced filtration* on X is the filtration with $F^nX = X \cap F^nA$.
(6) Given a filtered object (A, F) and a surjection $\pi : A \to Y$ the *quotient filtration* is the filtration with $F^nY = \pi(F^nA)$.
(7) A filtration F on an object A is said to be *finite* if there exist n, m such that $F^nA = A$ and $F^mA = 0$.

(8) Given a filtered object (A, F) we say $\bigcap F^i A$ exists if there exists a biggest subobject of A contained in all $F^i A$. We say $\bigcup F^i A$ exists if there exists a smallest subobject of A containing all $F^i A$.
(9) The filtration on a filtered object (A, F) is said to be *separated* if $\bigcap F^i A = 0$ and *exhaustive* if $\bigcup F^i A = A$.

Definition 0.3. Let $\mathcal{A}$ be an abelian category. A complex $I^\bullet$ is *K-injective* if for every acyclic complex $M^\bullet$ we have $\mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet) = 0$.

Theorem 0.4. *Let $\mathcal{A}$ be a Grothendieck abelian category. For every complex $M^\bullet$ there exists a quasi-isomorphism $M^\bullet \to I^\bullet$ such that $M^n \to I^n$ is injective and I^n is an injective object of $\mathcal{A}$ for all n and $I^\bullet$ is a K-injective complex. Moreover, the construction is functorial in $M^\bullet$.*

Lemma 0.5. *Let $\mathcal{A}$ be a Grothendieck abelian category. Then*

(1) *$D(\mathcal{A})$ has both direct sums and products,*
(2) *direct sums are obtained by taking termwise direct sums of any complexes,*
(3) *products are obtained by taking termwise products of K-injective complexes.*

证明. Let $K_i^\bullet$, $i \in I$ be a family of objects of $D(\mathcal{A})$ indexed by a set I. We claim that the termwise direct sum $\bigoplus_{i\in I} K_i^\bullet$ is a direct sum in $D(\mathcal{A})$. Namely, let $I^\bullet$ be a K-injective complex. Then we have

$$\begin{aligned}\mathrm{Hom}_{D(\mathcal{A})}(\bigoplus\nolimits_{i\in I} K_i^\bullet, I^\bullet) &= \mathrm{Hom}_{K(\mathcal{A})}(\bigoplus\nolimits_{i\in I} K_i^\bullet, I^\bullet) \\ &= \prod\nolimits_{i\in I} \mathrm{Hom}_{K(\mathcal{A})}(K_i^\bullet, I^\bullet) \\ &= \prod\nolimits_{i\in I} \mathrm{Hom}_{D(\mathcal{A})}(K_i^\bullet, I^\bullet)\end{aligned}$$

as desired. This is sufficient since any complex can be represented by a K-injective complex by Theorem 0.4. To construct the product, choose a K-injective resolution $K_i^\bullet \to I_i^\bullet$ for each i. Then we claim that $\prod_{i\in I} I_i^\bullet$ is a product in $D(\mathcal{A})$. This follows from Derived Categories, Lemma lemma product K injective (Tag 0BK6). □

Lemma 0.6. *Let $\mathcal{A}$ be an abelian category. Let (K, L, M, f, g, h) be a distinguished triangle of $K(\mathcal{A})$. If two out of K, L, M are K-injective complexes, then the third is too.*

Primary problem: Tag 0BKI

Lemma 0.7. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let $K^\bullet$ be a filtered complex of $\mathcal{A}$, see Homology, Definition 0.1. Then there exists a morphism $j : K^\bullet \to J^\bullet$ of filtered complexes of $\mathcal{A}$ such that*

(1) *J^n, $F^p J^n$, $J^n/F^p J^n$ and $F^p J^n/F^{p'} J^n$ are injective objects of $\mathcal{A}$,*
(2) *$J^\bullet$, $F^p J^\bullet$, $J^\bullet/F^p J^\bullet$, and $F^p J^\bullet/F^{p'} J^\bullet$ are K-injective complexes,*
(3) *j induces quasi-isomorphisms $K^\bullet \to J^\bullet$, $F^p K^\bullet \to F^p J^\bullet$, $K^\bullet/F^p K^\bullet \to J^\bullet/F^p J^\bullet$, and $F^p K^\bullet/F^{p'} K^\bullet \to F^p J^\bullet/F^{p'} J^\bullet$.*

证明. By Theorem 0.4 we obtain quasi-isomorphisms $i : K^\bullet \to I^\bullet$ and $i^p : F^p K^\bullet \to I^{p,\bullet}$ as well as commutative diagrams

$$\begin{array}{ccc} K^\bullet & \longleftarrow & F^p K^\bullet \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i^p} \\ I^\bullet & \xleftarrow{\alpha^p} & I^{p,\bullet} \end{array} \quad \text{and} \quad \begin{array}{ccc} F^{p'} K^\bullet & \longleftarrow & F^p K^\bullet \\ {\scriptstyle i^{p'}}\downarrow & & \downarrow{\scriptstyle i^p} \\ I^{p',\bullet} & \xleftarrow{\alpha^{pp'}} & I^{p,\bullet} \end{array} \quad \text{for } p' \leq p$$

such that $\alpha^p \circ \alpha^{p'p} = \alpha^{p'}$ and $\alpha^{p'p''} \circ \alpha^{pp'} = \alpha^{pp''}$. The problem is that the maps $\alpha^p : I^{p,\bullet} \to I^\bullet$ need not be injective. For each p we choose an injection $t^p : I^{p,\bullet} \to J^{p,\bullet}$ into an acyclic K-injective complex $J^{p,\bullet}$ whose terms are injective objects of $\mathcal{A}$ (first map to the cone on the identity and then use the theorem). Choose a map of complexes $s^p : I^\bullet \to J^{p,\bullet}$ such that the following diagram commutes

$$\begin{array}{ccc} K^\bullet & \longleftarrow & F^pK^\bullet \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i^p} \\ I^\bullet & & I^{p,\bullet} \\ & {\scriptstyle s^p}\searrow & \downarrow{\scriptstyle t^p} \\ & & J^{p,\bullet} \end{array}$$

This is possible: the composition $F^pK^\bullet \to J^{p,\bullet}$ is homotopic to zero because $J^{p,\bullet}$ is acyclic and K-injective (Derived Categories, Lemma lemma K injective (Tag 070I)). Since the objects $J^{p,n-1}$ are injective and since $F^pK^n \to K^n \to I^n$ are injective morphisms, we can lift the maps $F^pK^n \to J^{p,n-1}$ giving the homotopy to a map $h^n : I^n \to J^{p,n-1}$. Then we set s^p equal to $h \circ \mathrm{d} + \mathrm{d} \circ h$. (Warning: It will not be the case that $t^p = s^p \circ \alpha^p$, so we have to be careful not to use this below.)

Consider

$$J^\bullet = I^\bullet \times \prod\nolimits_p J^{p,\bullet}$$

Because products in $D(\mathcal{A})$ are given by taking products of K-injective complexes (Lemma 0.5) and since $J^{p,\bullet}$ is isomorphic to 0 in $D(\mathcal{A})$ we see that $J^\bullet \to I^\bullet$ is an isomorphism in $D(\mathcal{A})$. Consider the map

$$j = i \times (s^p \circ i)_{p \in \mathbf{Z}} : K^\bullet \longrightarrow I^\bullet \times \prod\nolimits_p J^{p,\bullet} = J^\bullet$$

By our remarks above this is a quasi-isomorphism. It is also injective. For $p \in \mathbf{Z}$ we let $F^pJ^\bullet \subset J^\bullet$ be

$$\mathrm{Im}\left(\alpha^p \times (t^{p'} \circ \alpha^{pp'})_{p' \leq p} : I^{p,\bullet} \to I^\bullet \times \prod\nolimits_{p' \leq p} J^{p',\bullet}\right) \times \prod\nolimits_{p' > p} J^{p',\bullet}$$

This complex is isomorphic to the complex $I^{p,\bullet} \times \prod_{p'>p} J^{p,\bullet}$ as $\alpha^{pp} = \mathrm{id}$ and t^p is injective. Hence $F^pJ^\bullet$ is quasi-isomorphic to $I^{p,\bullet}$ (argue as above). We have $j(F^pK^\bullet) \subset F^pJ^\bullet$ because of the commutativity of the diagram above. The corresponding map of complexes $F^pK^\bullet \to F^pJ^\bullet$ is a quasi-isomorphism by what we just said. Finally, to see that $F^{p+1}J^\bullet \subset F^pJ^\bullet$ use that $\alpha^{p+1p} \circ \alpha^{pp'} = \alpha^{p+1p'}$ and the commutativity of the first displayed diagram in the first paragraph of the proof.

We claim that $j : K^\bullet \to J^\bullet$ is a solution to the problem posed by the lemma. Namely, F^pJ^n is an injective object of $\mathcal{A}$ because it is isomorphic to $I^{p,n} \times \prod_{p'>p} J^{p',n}$ and products of injectives are injective. Then the injective map $F^pJ^n \to J^n$ splits and hence the quotient J^n/F^pJ^n is injective as well as a direct summand of the injective object J^n. Similarly for $F^pJ^n/F^{p'}J^n$. This in particular means that $0 \to F^pJ^\bullet \to J^\bullet \to J^\bullet/F^pJ^\bullet \to 0$ is a termwise split short exact sequence of complexes, hence defines a distinguished triangle in $K(\mathcal{A})$ by fiat. Since $J^\bullet$ and $F^pJ^\bullet$ are K-injective complexes we see that the same is true for $J^\bullet/F^pJ^\bullet$ by Derived Categories, Lemma 0.6. A similar argument shows that $F^pJ^\bullet/F^{p'}J^\bullet$ is K-injective. By construction $j : K^\bullet \to J^\bullet$ and the induced maps $F^pK^\bullet \to F^pJ^\bullet$ are quasi-isomorphisms. Using the long exact cohomology sequences of the complexes in play we find that the same holds for $K^\bullet/F^pK^\bullet \to J^\bullet/F^pJ^\bullet$ and $F^pK^\bullet/F^{p'}K^\bullet \to F^pJ^\bullet/F^{p'}J^\bullet$. $\square$